

LECTURE 7

GROUP AND RING THEORY

Definition 5.17. An R -module M is called a *free module* if M admits a basis.

In other words, the module M is free if there exists a subset S of M such that M is generated by S , and S is a linearly independent set. Clearly all vector spaces are free modules.

Example 5.18. (i) The ring R is a free module over itself. Any invertible element is a basis.

(ii) If R is commutative, the polynomial ring $R[X]$ in the indeterminate X is a free module with a basis $\{1, X, X^2, \dots\}$.

(iii) We regard the zero module (0) as a free module whose basis is the empty set. Indeed, recall that $R \neq 0$, and hence there is $0 \neq r \in R$. Therefore if a basis of (0) was non-empty, i.e. if the basis was $\{0\}$, then we would have a linear dependence relation $r0 = 0$, which contradicts the definition of a basis.

Theorem 5.19. Let M be a free R -module with a basis $\{f_1, \dots, f_n\}$. Then $M \cong R^n$ as R -modules.

Proof. Define a mapping $\phi: M \rightarrow R^n$ by

$$\phi\left(\sum_{i=1}^n r_i f_i\right) = \sum_{i=1}^n r_i e_i,$$

where $r_i \in R$ and

$$e_i = (0, \dots, 1, 0, \dots, 0) \in R^n.$$

Since $\sum_{i=1}^n r_i f_i = \sum_{i=1}^n r'_i f_i$ implies, by the linear independence of the f_i 's, that $r_i = r'_i$ for all i , the mapping ϕ is well defined. If $m = \sum_{i=1}^n r_i f_i$, $m' = \sum_{i=1}^n r'_i f_i$, and $r \in R$, then it immediately follows that

$$\phi(m + m') = \phi(m) + \phi(m') \quad \text{and} \quad \phi(rm) = r\phi(m).$$

Further, if $\phi(m) = \mathbf{0}$, then $\sum_{i=1}^n r_i e_i = \mathbf{0}$. This implies that the n -tuple $(r_1, \dots, r_n) = \mathbf{0}$, and hence each $r_i = 0$, proving that ϕ is one-to-one. It is clear that ϕ is onto, and hence ϕ is an R -isomorphism. $\square$

Theorem 5.20. Let M be a finitely generated free module over a ring R . Then all bases of M are finite.

Proof. Let (e_i) , for $i \in \Lambda$, be a basis of M , and let $\{x_1, \dots, x_n\}$ be a set of generators of M . Then each x_j can be written as

$$x_j = \sum_i \alpha_{ij} e_i, \quad \text{for } \alpha_{ij} \in R,$$

and all but a finite number of the α_{ij} are zero. Thus, the set S of those e_i 's that occur in the expressions of all the x_j 's, for $j = 1, 2, \dots, n$, is finite. $\square$

Theorem 5.21. Let M be a finitely generated free module over a commutative ring R . Then all bases of M have the same number of elements.

Proof. Equivalently, the theorem states that if $R^m \cong R^n$, then $m = n$. Let $m < n$, let $\phi : R^m \rightarrow R^n$ be an R -isomorphism, and let $\psi = \phi^{-1}$. Let $(e_1, \dots, e_m)$ and $(f_1, \dots, f_n)$ be ordered bases of R^m and R^n , respectively. Let us write

$$\begin{aligned}\phi(e_i) &= a_{1i}f_1 + \dots + a_{ni}f_n, \quad \text{for } 1 \leq i \leq m, \\ \psi(f_j) &= b_{1j}e_1 + \dots + b_{mj}e_m, \quad \text{for } 1 \leq j \leq n.\end{aligned}$$

Let $A = (a_{ji})$ and $B = (b_{kj})$ be $n \times m$ and $m \times n$ matrices. Then, for $1 \leq i \leq m$,

$$\psi\phi(e_i) = \psi\left(\sum_{j=1}^n a_{ji}f_j\right) = \sum_{k=1}^m \sum_{j=1}^n b_{kj}a_{ji}e_k.$$

Thus, by the linear independence of the e_i 's and by the fact that $\psi = \phi^{-1}$ we have

$$\sum_{j=1}^n b_{kj}a_{ji} = \delta_{ki}.$$

This yields

$$BA = \begin{pmatrix} b_{11} & \dots & b_{1n} \\ \vdots & & \vdots \\ b_{m1} & \dots & b_{mn} \end{pmatrix} \begin{pmatrix} a_{11} & \dots & a_{1m} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nm} \end{pmatrix} = I_m,$$

where I_m is the $m \times m$ identity matrix. Similarly $AB = I_n$. Let

$$A' = \begin{pmatrix} A & 0 \end{pmatrix} \quad \text{and} \quad B' = \begin{pmatrix} B \\ 0 \end{pmatrix}$$

be $n \times n$ augmented matrices, where each of the 0 blocks is a matrix of appropriate size. Then

$$A'B' = I_n, \quad \text{and} \quad B'A' = \begin{pmatrix} I_m & 0 \\ 0 & 0 \end{pmatrix}.$$

This implies $\det(A'B') = 1$ and $\det(B'A') = 0$. However A' and B' are $n \times n$ matrices over a commutative ring. So $\det(A'B') = \det(A')\det(B') = \det(B'A')$, which yields a contradiction. Hence $m \geq n$. By symmetry $n \geq m$. This proves $m = n$.

An alternative shorter approach is to just consider $AB = I_n = A'B'$ and the fact that $1 = \det(A'B') = \det(A')\det(B') = 0 \cdot 0 = 0$, a contradiction. $\square$

We remark that the multiplicative property of determinants, for matrices in $M_n(R)$, fails if R is not commutative. For example, consider the ring $\mathbb{H}$ of quaternions

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}, i^2 = j^2 = k^2 = -1, ij = k, ji = -k\}.$$

A counterexample is given by

$$A = \begin{pmatrix} i & 0 \\ 0 & j \end{pmatrix},$$

as $\det A = k$ and $(\det A)^2 = -1$ but $\det(A^2) = \begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix} = 1$.

Definition 5.22. The number of elements in any basis of a finitely generated free module M over a commutative ring R with unity is called the *rank* of M , written $\text{rank } M$ (or $\text{rk}(M)$).

In particular, if R is a field then the rank of M is known as the *dimension* of the vector space M and is denoted by $\dim M$ (or $\dim(M)$).

Example 5.23. (i) Let R be a commutative ring with unity. Then R is a free module of rank one over itself. The free module R^n has rank n .

(ii) The zero module (0) has rank zero.

Lemma 5.24. *Every finitely generated R -module, where R is with unity, is a homomorphic image of a finitely generated free module.*

Proof. Let M be an R -module with generators $x_1, \dots, x_n$. Let e_i be the n -tuple with all entries 0 except at the i th place, where the entry is 1. Then $e_1, \dots, e_n$ are linearly independent over R and generate a free module R^n .

Define a mapping $\phi : R^n \rightarrow M$ by

$$\phi \left(\sum_{i=1}^n r_i e_i \right) = \sum_{i=1}^n r_i x_i.$$

Since each element $x \in R^n$ has a unique representation as $\sum_{i=1}^n r_i e_i$, the mapping ϕ is well defined. Further, if $x = \sum_{i=1}^n r_i e_i$, $y = \sum_{i=1}^n r'_i e_i$, and $r \in R$, then it is clear that

$$\phi(x + y) = \phi(x) + \phi(y), \quad \phi(rx) = r\phi(x).$$

Hence ϕ is an R -homomorphism of R^n onto M . If K is the kernel of ϕ , it follows by the First Isomorphism Theorem that $R^n/K \cong M$. $\square$

In general, a submodule of a free R -module may not be free (see Exercise Sheet 7). But if R is a PID, we have the following.

Lemma 5.25. *Let R be a principal ideal domain, and let F be a free R -module with a basis consisting of n elements. Then any submodule K of F is also free with a basis consisting of m elements, such that $m \leq n$.*

Proof. We know that $F \cong R^n$ as R -modules (Theorem 5.19). To prove the lemma, we use induction on n . Also, here R^0 is interpreted as the trivial module (0) , and this is free on the empty set. Therefore, we may assume that $n > 0$, and let us identify the copy of K in R^n (under the isomorphism $F \cong R^n$) with K itself. Let $\pi : K \rightarrow R$ be the R -homomorphism defined by

$$\pi((x_1, \dots, x_n)) = x_1.$$

If $\pi = 0$, then $K = \ker \pi = \{(0, x_2, \dots, x_n) \mid x_i \in R\} \cap K \hookrightarrow R^{n-1}$, and the lemma follows by induction on n . If $\pi \neq 0$, its image is a non-zero ideal $(a) = Ra$ in R ; i.e. $\pi(K) = Ra$ for $a \neq 0$, since R is a PID. Choose $k \in K$ such that $\pi(k) = a$. We assert that

$$K = Rk \oplus \ker \pi.$$

For, let $x \in K$. Write $\pi(x) = ba$ for $b \in R$. Then $\pi(x - bk) = ba - ba = 0$. Hence

$$x = bk + (x - bk)$$

implies that

$$x \in Rk + \ker \pi.$$

Thus,

$$K = Rk + \ker \pi.$$

To prove that the sum is direct, let $ck \in Rk \cap \ker \pi$ with $c \in R$. Then $0 = \pi(ck) = ca$. Hence $c = 0$ because $a \neq 0$, as R is a free module. This proves our assertion that $K = Rk \oplus \ker \pi$.

It is easy to check that the mapping $r \mapsto rk$ of R onto Rk is an R -isomorphism. Also recall $\ker \pi \hookrightarrow R^{n-1}$. Hence by induction $\ker \pi$ is free, with a basis consisting of at most $n-1$ generators. Therefore,

$$K = \ker \pi \oplus Rk \cong R^s \oplus R, \quad \text{for some } s \leq n-1,$$

so

$$K \cong R^m, \quad \text{for } m := s+1 \leq n. \quad \square$$

6. NOETHERIAN AND ARTINIAN MODULES AND RINGS

6.1. Noetherian and Artinian modules. Here we consider two families of modules with similar finiteness properties that occur naturally in many settings.

Definition 6.1. An R -module M is called *Noetherian* (respectively *Artinian*) if for every ascending (respectively descending) sequence of R -submodules of M ,

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots \quad (M_1 \supseteq M_2 \supseteq M_3 \supseteq \cdots),$$

there exists a positive integer k such that

$$M_k = M_{k+1} = M_{k+2} = \cdots.$$

If M is Noetherian (respectively Artinian), then we also say that the *ascending* (respectively *descending*) *chain condition* for submodules holds in M , or M has acc (dcc) on submodules, or simply that M has ACC (respectively DCC).

Example 6.2. All finite modules are both Noetherian and Artinian, since they have only a finite number of submodules. For example, the $\mathbb{Z}$ -module $\mathbb{Z}/n\mathbb{Z}$.

Example 6.3. As the ring of integers $\mathbb{Z}$ is a principal ideal domain, any ascending chain of ideals of $\mathbb{Z}$ is of the form

$$(n) \subseteq (n_1) \subseteq (n_2) \subseteq \cdots,$$

where $n, n_1, n_2, \dots$ are in $\mathbb{Z}$. Since $(n_i) \subseteq (n_{i+1})$ implies $n_{i+1} \mid n_i$, any ascending chain of ideals in $\mathbb{Z}$ starting with n can have at most n distinct terms. This shows that $\mathbb{Z}$ as a $\mathbb{Z}$ -module is Noetherian. However $\mathbb{Z}$ as a $\mathbb{Z}$ -module has an infinite properly descending chain

$$(n) \supseteq (n^2) \supseteq (n^3) \supseteq \cdots,$$

showing that $\mathbb{Z}$ is not Artinian as a $\mathbb{Z}$ -module.

Example 6.4. The set $\mathbb{C}^*$ of non-zero complex numbers is an abelian group, and is also a $\mathbb{Z}$ -module. Consider the subset

$$U = \{x \in \mathbb{C}^* \mid x^{p^n} = 1 \text{ for some } n\}.$$

This is a subgroup of $\mathbb{C}^*$, hence is a $\mathbb{Z}$ -submodule.

Let $U_n = \{x \in \mathbb{C}^* \mid x^{p^n} = 1\}$ be the subgroup consisting of all p^n -th roots of unity. Then we have an ascending chain

$$U_1 \subsetneq U_2 \subsetneq U_3 \subsetneq \cdots$$

of submodules of U , which does not stabilise. So U is not a Noetherian $\mathbb{Z}$ -module.

However, one can show (exercise) that the U_i 's are the only submodules of U . Hence any descending chain stabilises.